

Sai Krishna Kanth Bandi

skkbandi01@gmail.com | (214) 308-8244 | Dallas, TX | [linkedin.com/in/saikrishnakanthbandi](https://www.linkedin.com/in/saikrishnakanthbandi)

Summary

Results-driven Java Developer with expertise in designing, developing, and deploying scalable, secure, and high-performance web applications. Backed by an ABET-accredited M.S. in Computer Science, with hands-on experience in Spring Boot, RESTful APIs, and modern MVC architectures. Proficient in object-oriented programming (OOP), Agile/Scrum methodologies, and software engineering best practices to deliver clean, maintainable, and testable code. Skilled in backend development, with a proven ability to collaborate across teams, analyze complex problems, and implement innovative solutions that drive business value.

Technical Skills

- **Programming:** Java, Python, C, TypeScript
- **Dev Tools:** IntelliJ, VS Code, Eclipse
- **Databases:** PostgreSQL, MySQL, SQL Server, Oracle
- **Build Tools:** Maven, Gradle
- **Cloud & DevOps:** AWS (EC2, RDS, S3, IAM), Docker, Kubernetes (EKS), Jenkins
- **Version Control:** Git (GitHub, GitLab, Bitbucket)
- **API Tools:** Postman, Swagger
- **Testing:** JUnit5, Mockito

Professional Experience

Software Developer, HealthChain

06/2025 – Present | Remote, USA

- Engineered and maintained a high-performance Java-based Payer Directory Service leveraging Vert.x, PostgreSQL, and modular microservices, achieving scalable and fault-tolerant architecture.
- Developed and secured RESTful APIs for authentication, payer/endpoint management, certificate lifecycle automation, and user operations, ensuring compliance and data integrity.
- Designed and deployed comprehensive activity logging, auditing, and monitoring frameworks, reducing incident investigation time by 40% and strengthening regulatory compliance.
- Automated certificate expiration alerts and revocation workflows using Quartz Scheduler and SMTP notifications, minimizing downtime incidents by 30%.
- Applied Java Bean Validation and centralized exception handling to enforce strict input validation, improving system resilience and error traceability.
- Containerized services with Docker and deployed on AWS ECS/RDS via CI/CD pipelines using Maven and GitHub Actions, enabling seamless rollbacks and faster releases.
- Utilized JSON-based configuration management and AWS Secrets Manager for environment-specific settings and secure secret handling.
- Built automated testing frameworks with JUnit, Mockito, and Selenium, achieving 95% coverage and reducing release defects from 50 to 5 per cycle.
- Architected a modular storage abstraction layer, enabling seamless integration with multiple storage backends and supporting scalable document operations.
- Collaborated in cross-functional Agile teams, contributing to design reviews, mentoring juniors, and delivering maintainable, high-quality microservices.

Student Intern, Cisco Networking Academy (AICTE)

03/2022 – 08/2022 | India

- Designed and delivered an IP-based access control system with Spring Boot, Java, and Hibernate ORM, enforcing restrictions for 500+ client IPs across multi-environment deployments.
- Developed REST controllers with custom security logic, improving modularity and ensuring consistent enforcement of access policies.
- Built robust IP filtering and validation mechanisms, reducing unauthorized access incidents by 90% and enhancing audit accuracy.
- Integrated MySQL with Hibernate ORM using JPA annotations, eliminating boilerplate code and improving database mapping efficiency by 30%.
- Implemented a DTO-driven architecture to optimize data encapsulation and streamline API payloads, reducing response size by 20%.
- Applied centralized exception handling with HandlerExceptionResolver, increasing error traceability and reducing runtime failures in production.

Projects

Employee Management System

- Designed and developed a scalable Java application using Spring Boot with embedded Tomcat and auto-configuration.
- Leveraged Spring MVC, REST, AOP, Cloud, and Data JPA to orchestrate microservices and streamline integration.
- Implemented full CRUD operations with MySQL via Hibernate ORM, optimizing query performance and reducing latency.
- Created secure DTOs to decouple service and controller layers, enhancing maintainability and minimizing payloads.
- Applied design patterns (Iterator, Abstract Factory, Prototype) to standardize code modularity and promote reusable components.

Certificate and Publication

- OCP 1Z0-829 Oracle Certified Professional: Java SE 17 Developer
- A Machine Learning-Powered Web Application for Accident Severity Detection, IEEE 2023

Education

University of North Texas - Denton, TX, USA

M.S. Computer Science

07/2023 – 05/2025